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Aeronautical cryogenic tank device for hydrogen storage, for external transportation by an aircraft

Abstract

Dismountable aeronautical pod device **30** with cryogenic storage capacity, for external transportation by an aircraft, comprising a front cryogenic tank **36**, a rear cryogenic tank **37**, and at least one central temporary storage tank for the rise in pressure of the gas supplied by the front cryogenic tank **36** and the rear cryogenic tank **37**.

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Background/Summary

(1) This application is the § 371 U.S. National Stage of International Application No. PCT/FR2023/050320, filed Mar. 9, 2023, which claims the priority to French Patent Application

No. FR2202234, filed Mar. 14, 2022.

(2) The present invention relates to the aeronautical industry.

(3) Since its inception, the aeronautical industry has used high-octane petrol engines. After 1945, the development of the jet engine and turbine led to the use of kerosene, which has a higher molecular weight than petrol, higher energy density, higher efficiency and lower flammability. These fuels are generally stored in tanks located in the wings, in the fuselage-wing link or in the tail.

(4) The trend towards reducing carbon dioxide emissions has led to engines that consume less fuel. However, the gains in carbon dioxide emissions are diminishing as certain technologies mature, in particular the speed of the blade tips. It became increasingly desirable to make a break with the past.

(5) This led to the development of gas-powered aircraft. Combustion of gases with a short or non-existent carbon chain, with oxygen where appropriate, results in little or no pollution. On the other hand, the storage of H₂, O₂ or C₁ or C₂ gases, due to the small size of the gas molecule, is difficult and liable to leaks.

(6) On the ground, such gases are generally stored in pressurised casings that are too heavy, too bulky and contain too much potential pressure energy to be carried on board an aircraft, or in welded and/or bonded cryogenic tanks. Cryopreservation of such gases is limited to a limited duration proportional to the stored volume.

(7) Furthermore, hydrogen, methane, ethane, ethylene, acetylene or oxygen stored in liquid state cannot be used by an internal or external combustion engine or a fuel cell. Final consumption requires a gaseous state.

(8) The need has arisen to store propulsion gas within an aircraft for on-board consumption, while applying aeronautical maintenance know-how and avoiding the need for new standardisation. Indeed, establishing new standards is a long, time-consuming process which risks delaying bringing gas-powered aircraft to market. Acquiring new maintenance know-how is also lengthy and costly, and can even lead to resistance.

(9) The invention proposes a dismountable aeronautical pod device with cryogenic storage capacity, for external transportation by an aircraft, comprising a front cryogenic tank, a rear cryogenic tank, at least one central temporary storage tank for the rise in pressure of the gas supplied by the front cryogenic tank and the rear cryogenic tank. Thanks to the invention, the device can have an elongated shape enabling it to be mounted in or on the aircraft with a high ratio of stored energy to device size. Storage and gasification take place separately and near one another. Dismountable means that the pod device can be mounted on and removed from an aircraft very quickly, on an airport runway, with or if possible without tools. It is advantageous that mounting is of the “plug and play” type.

(10) Unlike aerospace where parts are used once for a duration of a few seconds or tens of seconds, the aeronautical industry requires parts with a long life of several tens of thousands of hours and several tens of thousands of cycles.

(11) In one embodiment, each cryogenic tank, both the front and rear, has a convex outer shape, and comprises an inner casing defining a storage chamber, an outer casing containing the inner casing, an isolation chamber defined between the inner casing and the outer casing, and a removable manifold passing through the outer casing and the inner casing in a sealed manner, and a pipe fed by the manifold, and the central temporary storage tank having a quasi-toric or quasi-annular shape. The use of available space is improved.

(12) In one embodiment, each cryogenic tank is super-insulated under vacuum against conduction, convection and radiation. Natural evaporation is reduced.

(13) In one embodiment, the device is provided with a front cryogenic tank and a rear cryogenic tank, in an elongated shape along a common axis. An elongated outer shape of the device can be obtained.

- (14) In one embodiment, the device comprises at least two front cryogenic tanks and at least two rear cryogenic tanks, which are spherical in shape. The use of available space is improved for a contained mass.
- (15) In one embodiment, the device comprises a plurality of temporary storage tanks each arranged between two cryogenic tanks. The temporary storage capacity is increased.
- (16) In one embodiment, the central temporary storage tank forms a gasification member, an upstream valve being provided to be open for liquid to flow through during a filling phase of the central temporary storage tank and closed outside of the filling phase, a downstream valve being provided to be open for gas to flow through during an emptying phase of the central temporary storage tank and closed outside the emptying phase, the upstream valve and the downstream valve being closed during a gasification phase. The system ensures that the volume contained in the temporary tank(s) gives the aircraft the autonomy it needs, regardless of the state of the cryogenic tanks. The temporary tanks can be designed for a gas pressure of several hundred bar, although a selected gas pressure is supplied to the consumer members.
- (17) In one embodiment, the upstream valve and the downstream valve are controlled in a on-off manner. The valves are reliable.
- (18) In one embodiment, the device comprises a compressor arranged downstream of the downstream valve, said compressor being active at the end of the emptying phase to bring the pressure in the central temporary storage tank to a value lower than the lowest value of the pressure in the front cryogenic tank and in the rear cryogenic tank, and a pressure regulator arranged downstream of the downstream valve, said pressure regulator being active at the start of the emptying phase to bring the pressure of the gas at the outlet to a value lower than the pressure in the central temporary storage tank. The temporary tank can be emptied sufficiently so as to increase the amount of gas available for the consumer members and bring the temporary tank to a pressure at the end of emptying that is lower than the current pressure in the cryogenic tank designated at that time for filling. The temporary tank is filled by operating a cryogenic valve under the effect of the pressure difference. Dispensing with a cryogenic pump saves weight and reduces the risk of an incident.
- (19) In one embodiment, each of the cryogenic tanks is designed for an operating pressure of less than 8 bars and the central temporary storage tank is designed for an operating pressure of more than 500 bars. The cryogenic tanks have a reduced empty weight.
- (20) Other features and advantages of the invention will become apparent upon reading the detailed description below, and from the appended drawings, in which:

Description

- (1) FIG. 1 schematically shows an elevation side view of an aeronautical cryogenic tank device according to one aspect of the invention.
- (2) FIG. 2 schematically shows a longitudinal section of an aeronautical cryogenic tank device according to one aspect of the invention.
- (3) FIG. 3 schematically shows an exploded view of an aeronautical cryogenic tank device according to one aspect of the invention.
- (4) FIG. 4 schematically shows a longitudinal section of an aeronautical cryogenic tank device according to another aspect of the invention.
- (5) FIG. 5 schematically shows a longitudinal section of an aeronautical cryogenic tank device according to another aspect of the invention.
- (6) FIG. 6 schematically shows a distribution circuit according to one aspect of the invention.
- (7) The appended drawings may serve not only to complete the invention, but also to contribute to its definition, where appropriate.

- (8) The aeronautical gas storage device is designed to be transported by an aircraft: plane, drone, helicopter, etc. . . . The aeronautical gas storage device contains liquid and supplies gas. In other words, the gas is stored at a very low temperature in liquid form in a cryogenic tank. A cryogenic tank is unable to withstand high pressures, in particular greater than 10 bars.
- (9) Furthermore, the gas stored in liquid state cannot be used by an internal or external combustion engine or a fuel cell. Final consumption requires gas in a temperature range and pressure range specified by the manufacturer of the consumer member.
- (10) The stored gas is chosen from hydrogen, methane, ethane, ethylene, acetylene and oxygen.
- (11) The applicant also intends to take into account that gasification is a rapid phenomenon even in an ambient atmosphere at -55°C . found at altitude. By way of example, gaseous hydrogen at 0°C . and 1 atmosphere has a density approximately 800 times lower than liquid hydrogen at -253°C ., and therefore a volume approximately 800 times greater.
- (12) Furthermore, the rules of aeronautical maintenance require that most parts of the aircraft can be removed and repaired or replaced. In this way, an aircraft is able to land in any place—an aerodrome for a plane, a landing area for a helicopter—adapted to its weight and its landing requirements but not provided with maintenance equipment specific to the model of the aircraft. If damage is detected, the aircraft is designed to be repaired, either permanently or temporarily, or disassembled so as to replace or repair a faulty component, in line with the manufacturer's manuals and documents approved by the aviation safety authorities. It is desirable for the component to be readily accessible to a maintenance operator. If it is to be replaced, it is desirable for the component to be as small as possible for easy handling and transport. If it is to be repaired, it is desirable that the component can be repaired by tried-and-tested tools and methods commonly used in the aeronautical industry.
- (13) An aircraft undergoes daily, weekly, etc. inspections, grounding the aircraft for a duration inversely proportional to the frequency.
- (14) The applicant has identified a storage need, in particular for hydrogen, methane, ethane, ethylene, acetylene or oxygen, using aeronautical cryogenic tanks carried by the aircraft.
- (15) From another point of view, planes are currently subject to a rule stipulating maximum distance from a runway according to ETOPS certification. This distance depends on the type of plane.
- (16) Wishing to provide a high level of safety as well as for this safety to be noticed by users, the applicant has identified the need to fly even in the event of damage to the cryogenic tank requiring the contained gas to be released into the atmosphere.
- (17) The aeronautical storage pod device **30** aims to meet the complex need analysed in this way by the applicant.
- (18) The aeronautical cryogenic storage pod device **30** is designed to be transported by an aircraft. The aeronautical storage pod device **30** is filled with liquid and supplies gas at a selected pressure. In other words, the fuel or oxidiser is stored at a very low temperature in liquid form in a cryogenic tank. A cryogenic tank is unable to withstand high pressures, in particular greater than 10 bars.
- (19) The aeronautical storage device **30** is in the form of a pod. The aeronautical storage device **30** is fitted with a mechanism **31** for quick attachment to an aircraft wing.
- (20) In the embodiment shown in FIGS. **1** to **3**, the aeronautical cryogenic tank storage device **30** has an elongated shape. The aeronautical cryogenic tank storage device **30** comprises a central body **32**, a front end **33** and a rear end **34**. The front end **33** and the rear end **34** comprise shells to ensure an aerodynamic shape. The central body **32** comprises one or more generally cylindrical sections. In this case, three sections are provided, a central section **35**, a front section and a rear section. The shells and the section(s) of the central body can be removed. The shells and the section(s) of the main body provide mechanical protection against impacts, in particular bird strikes at the front end and handling shocks.
- (21) The attachment mechanism **31** is a quick-fit/disassembly mechanism enabling the aeronautical

cryogenic tank device **30** to be fitted to the aircraft in a short time, in particular a few minutes or tens of minutes, at the same time as other operations carried out on the runway. Said short time is less than the minimum turnaround time. The attachment mechanism **31** is provided substantially in the middle of the length of the aeronautical cryogenic tank device **30** or substantially close longitudinally to the centre of gravity of the aeronautical cryogenic tank device **30**. In this case, the attachment mechanism **31** is integral with the central section **35** of the central body. The attachment mechanism **31** comprises a gas line quick coupling. The central section **35** supports the front and rear sections. The front and rear sections support the front and rear ends respectively.

(22) As shown in FIG. 1, the central body **32**, the front end **33** and the rear end **34** have outer surfaces adapted to the flow of air at the speed of travel of the aircraft, in particular aerodynamic surfaces. Said outer surfaces form a fairing. Joints can be provided between the central body **32** and the front end **33**, on the one hand, and between the central body **32** and the rear end **34**, on the other. Vents are arranged in the central body **32**, the front end **33** and/or the rear end **34** for pressure balance and ventilation. Condensation and icing are prevented.

(23) The central body **32** provides support for the members arranged inside it and at its ends. The central body **32** can comprise at least one layer of shock-absorbing and vibration-absorbing material. The central body **32** acts as a shock-absorber limiting the stress on the attachment mechanism **31** and on the other members of the device described below. The central body **32** forms a self-supporting fairing.

(24) The aeronautical cryogenic tank device **30** comprises a front cryogenic tank **36** and a rear cryogenic tank **37**. The central section **35** supports the front cryogenic tank **36** and a rear cryogenic tank **37**.

(25) The front cryogenic tank **36** and a rear cryogenic tank **37** are mounted inside sections of the central body **32**. The front cryogenic tank **36** and the rear cryogenic tank **37** can have slightly different shapes in order to optimise the use of available space, whilst having a generally similar design. The front cryogenic tank and the rear cryogenic tanks are mounted head-to-tail. The front cryogenic tank **36** and the rear cryogenic tank **37** are shown here aligned along a longitudinal axis, this feature being optional.

(26) The front cryogenic tank **36** and the rear cryogenic tank **37** can have an elongated shape around a common axis or two axes. The front cryogenic tank **36** and the rear cryogenic tank **37** can have a curved front, a curved rear and a rotationally cylindrical central part.

(27) The front section substantially covers the front cryogenic tank **36**. The rear section substantially covers the rear cryogenic tank **37**. The central section **35** substantially covers the rear end of the front cryogenic tank **36**, the front end of the rear cryogenic tank **37** and a central space **38**. The central space **38** is generally ring-shaped. The central space **38** is at ambient pressure. The central space **38** is ventilated.

(28) The aeronautical cryogenic tank storage device **30** comprises at least one central temporary storage tank **7**. In this case, two central temporary storage tanks **7** are shown. The central temporary storage tanks **7** are mounted in parallel. The central temporary storage tanks **7** are managed by a distribution and conditioning circuit **1** shown in FIG. 6. The central temporary storage tanks **7** are alternatively filled with liquid coming from at least one of the cryogenic tanks **36** and **37** and, after gasification, emptied of the gas that they contain. The central temporary storage tanks **7** operate alternately, one filling, the other emptying and vice versa most of the operating time. It can be provided to take off with the two central temporary storage tanks **7** full.

(29) Each central temporary storage tank **7** forms a gasifier. Each central temporary storage tank **7** forms a heat exchanger. Each central temporary storage tank **7** comprises at least one cryogenic liquid inlet and at least one gas outlet. The central temporary storage tanks **7** do not have any thermal insulation. The central temporary storage tanks **7** comprise a simple casing. The central temporary storage tanks **7** are metallic and/or made of composite materials. The central temporary storage tanks **7** are made entirely or partly of conductive material. The central temporary storage

tanks **7** withstand cryogenic temperatures. The central temporary storage tanks **7** withstand high operating pressures compared with the low-pressure front **36** and rear **37** cryogenic tanks. The central temporary storage tanks **7** have a short storage life compared to the front **36** and rear **37** cryogenic tanks which have a long storage life.

(30) The central temporary storage tanks **7** are mounted in the central space, in this case opposite the quick attachment mechanism. In the embodiment shown in FIG. **2**, the central temporary storage tanks **7** have a cylindrical shape with a vertical axis and rounded ends.

(31) As the central space **38** is compact, the liquid lines and gas lines have a limited length. The weight of the device is optimised.

(32) In the embodiment shown in FIG. **4**, the aeronautical cryogenic tank comprises two front cryogenic tanks **36** and two rear cryogenic tanks **37**. The front cryogenic tanks **36** and the rear cryogenic tanks **37** have a spherical shape. The front cryogenic tanks **36** and the rear cryogenic tanks **37** are shown here aligned, this feature being optional. The central temporary storage tanks **7** have an annular, in particular toric shape. In addition, additional temporary storage tanks **39** are arranged between the front cryogenic tanks **36** and between the rear cryogenic tanks **37**. The additional cryogenic tanks **39** have an annular, in particular toric shape.

(33) Each cryogenic tank is insulated to contain liquid fuel or oxidiser at -253°C . Each cryogenic tank is able to withstand a maximum operating pressure of the order of 6 to 10 bars.

(34) In the embodiment shown in FIG. **5**, the additional buffer tanks **39** between the front cryogenic tanks **36** are arranged in the shape of a regular polygon. Each additional cryogenic tank **39** has a cylindrical body and rounded ends. The axes of the bodies of the additional buffer tanks **39** define a hexagon here.

(35) The additional buffer tanks **39** between the rear cryogenic tanks **37** are arranged in parallel. Each additional cryogenic tank **39** has a cylindrical body and rounded ends. The axis of the device and the axes of the additional buffer tanks **39** are parallel. In a cross-sectional view, the axes of the bodies of the additional buffer tanks **39** define the vertices of a regular polygon. The additional buffer tanks **39** are arranged like the chambers of a barrel.

(36) FIG. **5** shows two embodiments at the same time, the polygonal embodiment in the front and the parallel embodiment in the rear, grouped together for the sake of concision. In practice, a device has only one of these embodiments with either all of the additional buffer tanks **39** as a polygon, or all of the additional buffer tanks **39** in parallel. In the two embodiments, the design of the additional buffer tanks **39** is very robust owing to their shape adapted to high pressures and manufacturing constraints.

(37) In one embodiment, at least one additional cryogenic tank **39** is provided with a cylindrical body and rounded ends.

(38) The gas distribution circuit

(39) The distribution circuit **1** is explained with reference to FIG. **6**.

(40) The aircraft distribution circuit **1** is fed by the cryogenic tanks **36**, **37** to supply gas to one or more consumer members of the aircraft. A flowmeter **22** is arranged at the outlet of each cryogenic tank **2**.

(41) Each cryogenic tank is fitted with an outlet pipe **4**. The terms upstream and downstream refer to the direction of flow of the fluid, liquid or gas during normal operation.

(42) The distribution circuit **1** comprises a first valve **11** for each cryogenic tank. The first valve **11** is mounted on the outlet pipe **4**. The first valves **11** are controlled with an open position and a closed position. The intermediate positions of the first valves **11** are dynamic in the sense that the first valves **11** move as they pass through said intermediate positions. In other words, the first valves **11** are on-off. The first valves **11** can be arranged immediately downstream of the flowmeters **22**.

(43) The first valves **11** open into a cryogenic distributor **5**. The cryogenic distributor **5** can comprise a common pipe **6** connecting the outlets of the first valves **11**. The distributor is cryogenic

in the sense that liquid fuel/oxidiser passes through it.

(44) The cryogenic distributor 5 comprises a plurality of outlets, in this case three. Second valves 12 are mounted on each of said outlets. The second valves 12 are controlled with an open position and a closed position. The intermediate positions of the second valves 12 are dynamic in the sense that the second valves 12 move as they pass through said intermediate positions. In other words, the second valves 12 are on-off. There are three second valves 12 in this case.

(45) A central tank 7 is mounted downstream of each second valve 12. Three central tanks 7 are provided in this embodiment. Each central tank 7 also serves as a gasifier. Insulation can be prevented. Each central tank 7 receives liquid and supplies gas downstream. A pressure rise or gasification step takes place in each central tank 7 between filling and emptying. Each central tank 7 is able to withstand a maximum operating pressure of the order of 300 to 1000 bars. Each central tank 7 is designed to operate in a temperature range of -253°C. to $+60^{\circ}\text{C.}$ The central tanks 7 are two-phase for some of the operating steps and single-phase gas for the other operating steps. Each central tank 7 can be fitted with a heater 8.

(46) Downstream of each central tank 7 is a third valve 13 for supplying gas and a pressure regulator 9 downstream of the third valve 13. The pressure regulator 9 reduces the pressure to supply gas at a consumption pressure set by the manufacturer of the consumer member 3. The pressure regulator 9 is active when the pressure in the central tank 7 is greater than the consumption pressure and inactive otherwise. The consumption pressure is lower than the maximum pressure of the central tank 7. The consumption pressure is independent of the maximum pressure of the cryogenic tanks. The third valves 13 are on-off.

(47) A fourth controlled valve 14 can be provided downstream of each pressure regulator 9. The fourth valves 14 are on-off.

(48) The fourth valves 14 or the pressure regulators 9, depending on the chosen option, open into a manifold 10. The manifold 10 can comprise a pipe connecting the outlets of the fourth valves 14 or pressure regulators 9. Gas flows through the manifold 10. The manifold 10 is connected downstream to supply lines 23 to the consumer members 3. In general, a supply line 23 is provided for each consumer member 3. Each supply line 23 can be fitted with a controlled supply valve 24. The supply valve 24 has a variable flow rate.

(49) The distribution circuit 1 comprises at least one compressor 20 connected to the manifold 10. In general, two compressors 20 are provided in parallel for redundancy. The compressor 20 is electric. The compressor 20 can be fitted with a controlled upstream valve. The compressor 20 delivers gas to the manifold 10. In particular, the manifold 10 is composed of a pipe if there is a single consumer member 3.

(50) Downstream of each central tank 7 is a fifth valve 15 for supplying gas and a second manifold downstream of the fifth valves 15. The second manifold is connected to the compressor 20. The fifth valves 15 isolate the central tanks 7 and the compressor 20. The fifth valves 15 are controlled. The fifth valves 15 are on-off.

(51) The compressor 20 increases the pressure to supply gas at a pressure equal to a consumption pressure set by the manufacturer of the consumer member 3. The consumption pressure is lower than the maximum pressure in the central tank 7. The compressor 20 enables gas to be taken from a central tank 7, the pressure of which is lower than the consumption pressure, to supply the manifold 10 and the consumer members 3. More complete emptying of the central tank 7 increases the autonomy provided by the gas contained in the central tank 7 or reduces the volume of the central tank 7.

(52) Sufficient emptying of the central tank 7 to bring the internal pressure of the central tank 7 to a value lower than the pressure in one of the cryogenic tanks makes it possible to transfer the liquid from the cryogenic tank to the central tank 7 by pressure difference during filling that follows emptying. Thus, the liquid in the cryogenic tank is drawn into the central tank 7 until pressure equilibrium is reached. A cryogenic pump can be dispensed with, which results in weight savings

and a reduction in energy consumption.

(53) The distribution circuit **1** offers a combination of individual states for each cryogenic tank, for each central tank **7** and for each consumer member **3**. Several consumer members **3** can be active at the same time. In normal mode, one cryogenic tank is being emptied whilst the others are inactive and therefore closed. However, in certain situations, for example in order to reduce the pressure in several cryogenic tanks, one particular embodiment can be provided in which several cryogenic tanks are being emptied. The central tanks **7** have a filling mode, a gasification mode, a gas storage mode and an emptying mode.

(54) When one of the cryogenic tanks is being emptied, the first corresponding valve **11** is open and the other first valves **11** are closed. When one of the consumer members **3** is being supplied, the corresponding supply valve **24** is open.

(55) When one of the central tanks **7** is in filling mode, the second valve **12** connected to said central tank **7** is open and at least one of the first valves **11** is open. The other second valves **12** are closed expected in the event that two central tanks **7** are being filled simultaneously. The third valve **13** connected to said central tank **7** is closed. The fifth valve connected to said central tank **7** is closed.

(56) When one of the central tanks **7** is in gasification mode, the second valve **12** connected to said central tank **7**, the third valve **13** connected to said central tank **7** and the fifth valve **15** connected to said central tank **7** are closed. The gasification mode is brief, in particular when the ambient atmosphere is warm and/or the central tank **7** is heated.

(57) When one of the central tanks **7** is in emptying mode, the second valve **12** connected to said central tank **7** is closed. In the first emptying part, the pressure in the central tank **7** is greater than the consumption pressure. The third valve **13** connected to said central tank **7** is open, the fourth corresponding valve **14** is open and the fifth valve connected to said central tank **7** is closed. The gas undergoes a reduction in pressure in the pressure regulator **9** and is supplied to the manifold **10** at the consumption pressure. The gas is then consumed by the consumer member(s) **3**.

(58) At any given moment, out of the three central tanks **7**, one is in filling mode, another is in gasification then storage mode and the third is in emptying mode. As the modes have different durations, it is also possible to find two central tanks **7** in filling mode and the third in emptying mode or vice versa. It is also possible to find two central tanks **7** in storage mode and the third in emptying mode or vice versa.

(59) In the embodiment, a flowmeter **22** is arranged at the outlet of each liquid fuel/oxidiser source **2**. The flowmeters **22** provide sufficiently accurate information on the amount of liquid supplied to such a central tank **7**.

(60) In the embodiment, the distribution circuit **1** comprises a control unit **25** receiving an external command, for example from the consumer units **3** outside of the aeronautical storage device or from a central control unit of the aircraft, and liquid flow data from the flowmeters **22**. The control unit **25** generates and sends commands to said first, second, third, fourth and fifth controlled valves and to the controlled supply valves **24**. The commands can be “open” or “closed”. The control unit **25** manages said combination of individual states.

(61) Alternatively, the first valves **11** can be replaced by at least one multi-way valve having a plurality of inlets and one outlet. In this case, it is advantageous to provide a multi-way valve with mixed positions, in particular at least one position for simultaneously emptying two or more cryogenic tanks **2** to reduce the pressure while preventing loss into the atmosphere.

(62) Alternatively, the second valves **12** can be replaced by at least one multi-way valve having one inlet and a plurality of outlets, one for each central tank **7**. Said multi-way valve forms a distributor.

(63) Alternatively, the pressure regulators **9** are replaced by a single pressure regulator **9**, the third valves **13** opening to the single pressure regulator **9**. In this case, the third valves **13** can be replaced by at least one multi-way valve having a plurality of inlets and one outlet to the pressure regulator. The fourth valves **14** are therefore replaced by a single fourth valve **14**, where

appropriate not controlled.

(64) Alternatively, the fifth valves **15** can be replaced by at least one multi-way valve having a plurality of inlets, one for each central tank **7**, and one outlet to the compressor **20** or compressors **20**. Said multi-way valve forms a manifold **10**.

(65) As the cryogenic tanks **2** are subject to evaporation from the liquid, a gas collection circuit can be provided in an upper part of the cryogenic tanks **2**. The collection circuit can be activated above a threshold pressure by a calibrated pressure valve. The collection circuit comprises a compressor to re-inject the gas downstream, for example between the fifth valves **15** and the compressor **20**.

(66) Optionally, additional flowmeters are arranged at the inlet of each buffer tank. Liquid flow measurement redundancy is provided.

Claims

1. A dismountable aeronautical pod device with cryogenic storage capacity, for external transportation by an aircraft, comprising a front cryogenic tank, a rear cryogenic tank, at least one central gasification and temporary storage tank for a rise in pressure of gas supplied by the front cryogenic tank and the rear cryogenic tank, each cryogenic tank, both the front and rear, has a convex outer shape, and comprises an inner casing defining a storage chamber, an outer casing containing the inner casing, an isolation chamber defined between the inner casing and the outer casing, and a removable manifold passing through the outer casing and the inner casing in a sealed manner, and a pipe fed by the manifold, and the temporary storage tank having a quasi-toric or quasi-annular shape.
2. The device according to claim 1, wherein each cryogenic tank is super-insulated under vacuum against conduction, convection and radiation.
3. The device according to claim 1, provided with the front cryogenic tank and the rear cryogenic tank, in an elongated shape along a common axis.
4. The device according to claim 1, comprising at least two front cryogenic tanks and at least two rear cryogenic tanks, which are spherical in shape.
5. The device according to claim 1, comprising a plurality of temporary storage tanks each arranged between two cryogenic tanks.
6. The device according to claim 1, wherein the temporary storage tank forms a gasification member, an upstream valve being provided to be open for liquid to flow through during a filling phase of the temporary storage tank and closed outside of the filling phase, a downstream valve being provided to be open for gas to flow through during an emptying phase of the temporary storage tank and closed outside the emptying phase, the upstream valve and the downstream valve being closed during a gasification phase.
7. The device according to claim 6, wherein the upstream valve and the downstream valve are controlled in an on-off manner.
8. The device according to claim 1, wherein each of the cryogenic tanks is designed for an operating pressure of less than 8 bars and the temporary storage tank is designed for an operating pressure of more than 500 bars.
9. A dismountable aeronautical pod device with cryogenic storage capacity, for external transportation by an aircraft, comprising a front cryogenic tank, a rear cryogenic tank, at least one central gasification and temporary storage tank for a rise in pressure of gas supplied by the front cryogenic tank and the rear cryogenic tank, comprising a compressor arranged downstream of a downstream valve, said compressor being active at an end of an emptying phase to bring the pressure in the temporary storage tank to a value lower than a lowest value of the pressure in the front cryogenic tank and in the rear cryogenic tank, and a pressure regulator arranged downstream of the downstream valve, said pressure regulator being active at a start of the emptying phase to bring the pressure of the gas at an outlet to a value lower than the pressure in the temporary storage

tank.

10. The device according to claim 9, wherein each of the cryogenic tanks is designed for an operating pressure of less than 8 bars and the temporary storage tank is designed for an operating pressure of more than 500 bars.

11. The device according to claim 9, wherein each cryogenic tank is super-insulated under vacuum against conduction, convection and radiation.

12. The device according to claim 9, provided with the front cryogenic tank and the rear cryogenic tank, in an elongated shape along a common axis.

13. The device according to claim 9, comprising at least two front cryogenic tanks and at least two rear cryogenic tanks, which are spherical in shape.

14. The device according to claim 9, comprising a plurality of temporary storage tanks each arranged between two cryogenic tanks.

15. The device according to claim 9, wherein the temporary storage tank forms a gasification member, an upstream valve being provided to be open for liquid to flow through during a filling phase of the temporary storage tank and closed outside of the filling phase, a downstream valve being provided to be open for gas to flow through during an emptying phase of the temporary storage tank and closed outside the emptying phase, the upstream valve and the downstream valve being closed during a gasification phase.
